Henry Tran

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Education

CUNY City College of New York *New York City, NY*
Bachelor of Science in Computer Science *June 2028*
Coursework: Software Engineering, Linear Algebra, Algorithms, Discrete Math, Data Structures, Software Design, Calc III, Physics II, Probability & Statistics for CS

Professional Experience

Kapitus *New York City, NY*
Financial Data Engineer Intern *January - May 2026*

- Refactored per-facility Alteryx macros into one master macro, cutting daily ETL runtime by 75% and speeding up loan portfolio reporting to investors
- Consolidated facility tables into a unified MySQL view as a single source of truth for investor reporting
- Built automated data quality checks to catch anomalies before loan data reached institutional investors
- Documented the consolidated macro logic and MySQL view so the team could maintain the workflow

MIT Museum Maker Hub *Cambridge, MA*
Maker & Facilitator *November 2024*

- Engineered and fabricated a custom circuit board selected by museum staff to be displayed as an educational exemplar for visiting youth
- Mentored K-12 students during drop-in STEAM activities, guiding them through basic electronics, troubleshooting, and design principles

Projects

Echo | Python, FastAPI, PostgreSQL, SQLAlchemy, SciPy *October 2026*

- Built and deployed a full-stack web app (echo.henrytran.dev) that matches Spotify listeners who share core taste and can introduce each other to new artists
- Engineered a recommendation engine using IDF-weighted cosine similarity and co-listening scores on sparse user-artist matrices, with two-stage retrieval and explainable suggestions
- Designed an idempotent ETL pipeline from the Spotify API to PostgreSQL with OAuth, async fetching, exponential-backoff retries, and artist entity resolution

Organizational Experience

Tutoring *August 2023 - May 2024*

- **AP Tutor** - Taught students AP courses, helping them achieve class averages of 95%+
- **SHSAT Tutor** - Tutored students into specialized schools such as The Bronx High School of Science

Robotics Team *August - December 2024*

- **Micromouse** - Developed an autonomous maze-solving robot, writing C++ firmware for a flood-fill pathfinding algorithm and PID motor control

Skills

Tools: Python, SQL, PostgreSQL, MySQL, Alteryx, FastAPI, SQLAlchemy, NumPy/SciPy, Git, C++, Java, JavaScript, R
Languages: English, Vietnamese
Interests: Data Engineering | Cloud Engineering | AI/ML | Product Management | Astronomy | Finance